Name :Aditi Bhoir

124A7004

B3

EXPERIMENT NO. 3

IMPLEMENTATION OF STATIC ROUTING USING CISCO PACKET TRACER

EXPERIMENT NO. 03

AIM: - To implement static routing using Cisco Packet Tracer.

OBJECTIVES:

a. **To create routing topology.**

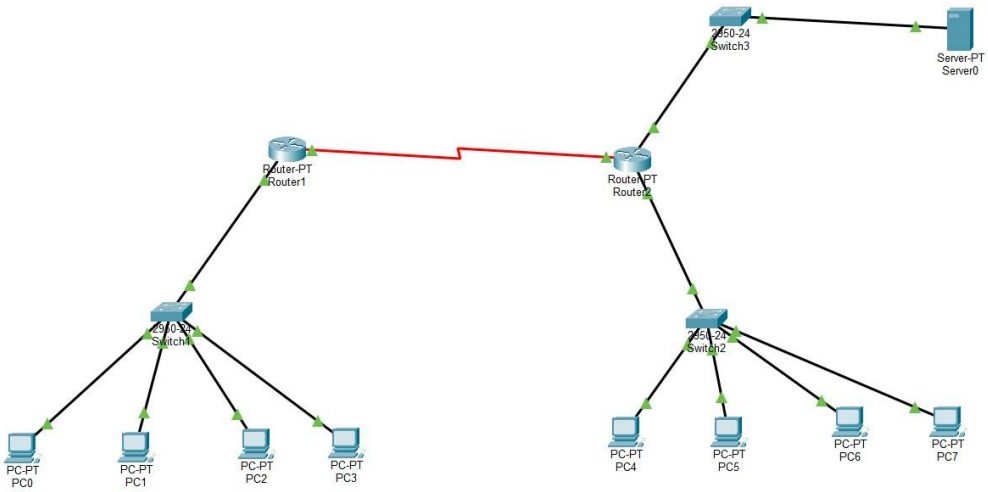
b. **To configure IP addresses.**

c. **To define static routes.**

d. **To check the connectivity among the devices in the network.**

SOFTWARE: Cisco Packet Tracer 7.3.0.

SCENARIO:



CONFIGURATION OF SERIAL2/0 INTERFACE OF Router0

```
Router>en
Router#config t
Router(config)#in se
Router(config)#in serial 2/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shut
```

CONFIGURATION OF FASTETHERNET0/0 INTERFACE OF Router0

```
Router(config)#in fa
Router(config)#in fastEthernet 0/0
Router(config-if)#ip address 172.0.0.5 255.255.0.0
Router(config-if)#no shut
```

CONFIGURATION OF FASTETHERNET0/0 INTERFACE OF Router1

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#in fa 0/0
Router(config-if)#ip address 192.168.0.5 255.255.255.0
Router(config-if)#no shut
```

CONFIGURATION OF SERIAL2/0 INTERFACE OF Router1

```
Router(config)#in se
Router(config)#in serial 2/0
Router(config-if)#ip address 10.0.0.2 255.0.0.0
Router(config-if)#no shut
```

CONFIGURATION OF FASTETHERNET1/0 INTERFACE OF Router1

```
Router>en
Router#ipiiniconfig t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#fin fa
Router(config)#in fastEthernet 1/0
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#no shut
```

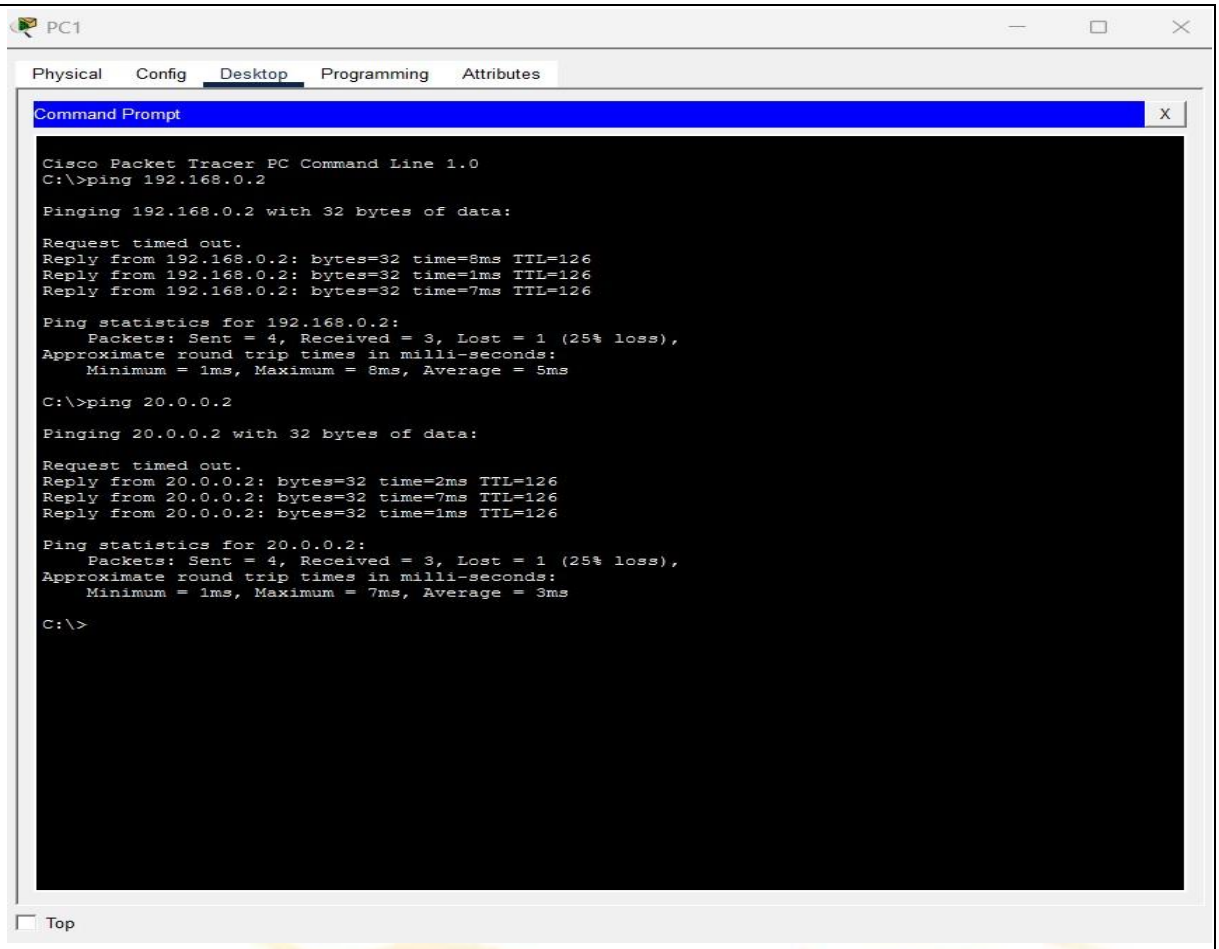
STATIC ROUTE Router0

```
Router(config-if)#exit
Router(config)#ip route 192.168.0.0 255.255.255.0 10.0.0.2
Router(config)#ip route 20.0.0.0 255.0.0.0 10.0.0.2
Router(config)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
```

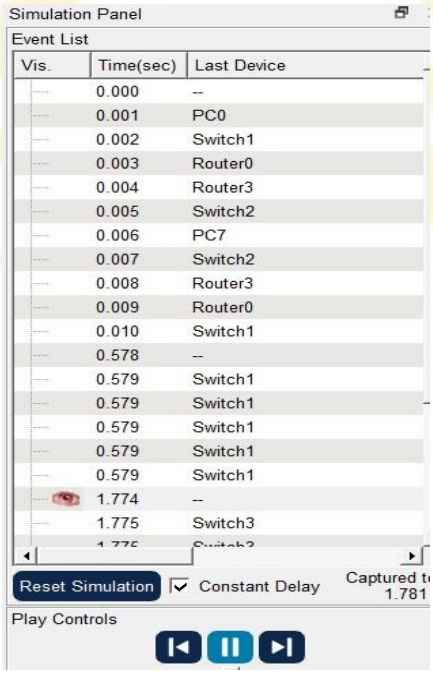
STATIC ROUTE Router1

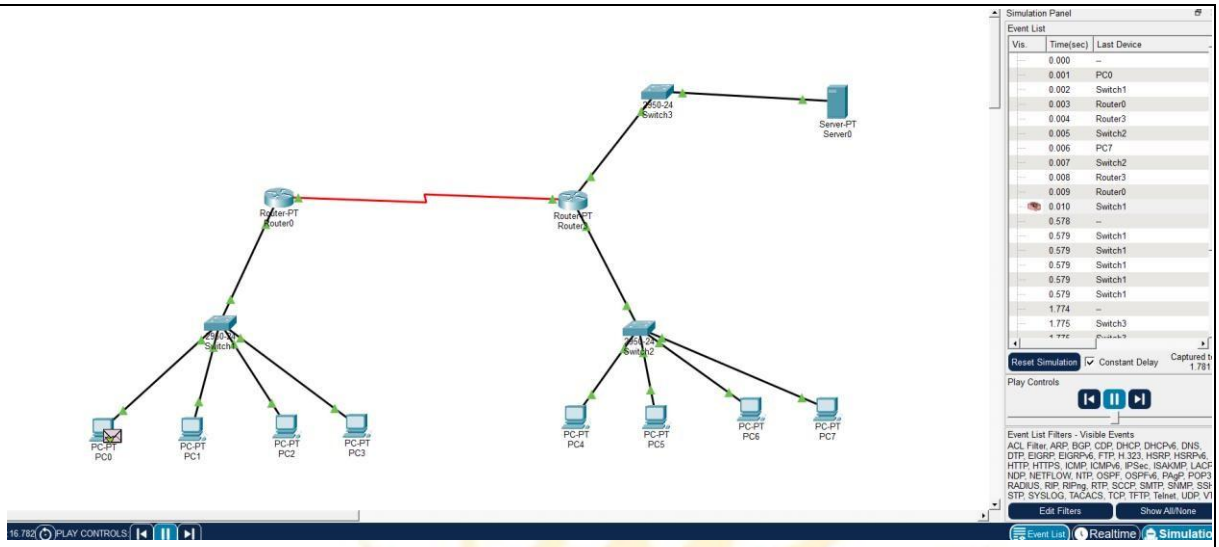
```
Router(config)#iniip route 172.0.0.0 255.255.0.0 10.0.0.2
%Invalid next hop address (it's this router)
Router(config)#ip route 172.0.0.0 255.255.0.0 10.0.0.1
```

PING RESPONSE:



PACKET FLOW SEQUENCE:





CONCLUSION:

A network topology was built in Cisco Packet Tracer, with IP addresses assigned to hosts and interfaces via CLI. Static routes were configured using `ip route`, and successful end-to-end connectivity was verified using `ping`.